data_in[7:0]

rst clock

wr

rd =wr_ptr1_i

RTL_EQ

I0

I1 O +wr_ptr0_i

RTL_ADD

I1

I0[4:0] O[4:0]

wr_ptr0_i__0 RTL_AND

I0

I1 O

wr_ptr_reg[4:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[0]_i
RTL_ROM

A[4:0] O

mem_reg[1]_i
RTL_ROM

A[4:0] O

mem_reg[2]_i
RTL_ROM

A[4:0] O

mem_reg[3]_i
RTL_ROM

A[4:0] O

mem_reg[4]_i
RTL_ROM

A[4:0] O

mem_reg[5]_i
RTL_ROM

A[4:0] O

mem_reg[6]_i
RTL_ROM

A[4:0] O

mem_reg[7]_i
RTL_ROM

A[4:0] O

mem_reg[8]_i
RTL_ROM

A[4:0] O

mem_reg[9]_i
RTL_ROM

A[4:0] O

mem_reg[10]_i
RTL_ROM

A[4:0] O

mem_reg[11]_i
RTL_ROM

A[4:0] O

mem_reg[12]_i
RTL_ROM

A[4:0] O

mem_reg[13]_i
RTL_ROM

A[4:0] O

mem_reg[14]_i
RTL_ROM

A[4:0] O

mem_reg[15]_i
RTL_ROM

A[4:0] O

mem_reg[16]_i
RTL_ROM

A[4:0] O

mem_reg[17]_i
RTL_ROM

A[4:0] O

mem_reg[18]_i
RTL_ROM

A[4:0] O

mem_reg[19]_i
RTL_ROM

A[4:0] O

mem_reg[20]_i
RTL_ROM

A[4:0] O

mem_reg[21]_i
RTL_ROM

A[4:0] O

mem_reg[22]_i
RTL_ROM

A[4:0] O

mem_reg[23]_i
RTL_ROM

A[4:0] O

mem_reg[24]_i
RTL_ROM

A[4:0] O

mem_reg[25]_i
RTL_ROM

A[4:0] O

mem_reg[26]_i
RTL_ROM

A[4:0] O

mem_reg[27]_i
RTL_ROM

A[4:0] O

mem_reg[28]_i
RTL_ROM

A[4:0] O

mem_reg[29]_i
RTL_ROM

A[4:0] O

mem_reg[30]_i
RTL_ROM

A[4:0] O

mem_reg[31]_i
RTL_ROM

A[4:0] O

mem_i__0

RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__1
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__2
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__3
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__4
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__5
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__6
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__7
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__8
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__9
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__10
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__11
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__12
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__13
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__14
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__15
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__16
RTL_MUX

S=1'b1 I0

S=default I1 O S

mem_i__17

RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__18
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__19
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__20
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__21
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__22
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__23
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__24
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__25
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__26
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__27
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__28
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__29
RTL_MUX

S=1'b1 I0

S=default I1 O S mem_i__30
RTL_MUX

S=1'b1 I0

S=default I1 O S

mem_i

RTL_MUX

S=1'b1 I0

S=default I1 O S+

rd_ptr0_i
RTL_ADD

I1

I0[4:0] O[4:0]

mem_reg[0][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[1][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[2][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[3][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[4][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[5][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[6][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[7][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[8][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[9][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[10][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[11][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[12][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[13][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[14][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[15][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[16][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[17][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[18][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[19][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[20][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[21][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[22][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[23][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[24][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[25][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[26][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[27][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[28][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[29][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[30][7:0]

RTL_REG_SYNC C

CED Q

RST

mem_reg[31][7:0]

RTL_REG_SYNC C

CED Q

RST

rd_ptr_reg[4:0]

RTL_REG_SYNC C

CED Q

RST =

data_out1_i
RTL_EQ

I0

I1 O

mem_i__31

RTL_MUX

S=5'b11111 I0[7:0] S=5'b11110 I1[7:0] S=5'b10101 I10[7:0] S=5'b10100 I11[7:0] S=5'b10011 I12[7:0] S=5'b10010 I13[7:0] S=5'b10001 I14[7:0] S=5'b10000 I15[7:0] S=5'b01111 I16[7:0] S=5'b01110 I17[7:0] S=5'b01101 I18[7:0] S=5'b01100 I19[7:0] S=5'b11101 I2[7:0] S=5'b01011 I20[7:0] S=5'b01010 I21[7:0] S=5'b01001 I22[7:0] S=5'b01000 I23[7:0] S=5'b00111 I24[7:0] S=5'b00110 I25[7:0] S=5'b00101 I26[7:0] S=5'b00100 I27[7:0] S=5'b00011 I28[7:0] S=5'b00010 I29[7:0] S=5'b11100 I3[7:0] S=5'b00001 I30[7:0] S=5'b00000 I31[7:0] S=5'b11011 I4[7:0] S=5'b11010 I5[7:0] S=5'b11001 I6[7:0] S=5'b11000 I7[7:0] S=5'b10111 I8[7:0] S=5'b10110 I9[7:0]

O[7:0]

S[4:0]-

empty0_i
RTL_SUB

I0[4:0]

I1[4:0] O[5:0]

data_out0_i RTL_AND

I0

I1 O

data_out_reg[7:0]

RTL_REG_SYNC C

CED Q

RST =

empty_i

RTL_EQO

I0[5:0]

I1[5:0] =

full_i

RTL_EQO

I0[5:0] V=B"011111" I1[5:0]

data_out[7:0]

full empty